

Design Description

DOMAIN, WHO, WHY:

Domain: Australian sports participation, attendance, and funding efficiency. Who:

General Australian public (18-60) interested in sports but not necessarily data experts.

Why: To reveal participation gaps by age/gender/state, show COVID-19 impact on attendance, analyse funding ROI, and suggest pathways to increase participation.

WHAT - DATA SOURCES:

Data aggregated from AusPlay Survey 2023-24, ASC Annual Reports, and sporting body annual reports. Some 2025 figures are projections based on Q1-Q2 actuals. Eight datasets synthesised from authoritative sources: AusPlay Survey (Australian Sports Commission), ABS Sport Participation, AFL/Cricket Australia annual reports, and Sport Australia State of Play. Each CSV includes: age_gender_sport_matrix (participation by age/gender/sport), attendance_streamgraph (2019-2025 attendance), dominant_sport_by_state (state-level preferences), funding_vs_medals (2019 vs 2025), participation_pathways (entry to outcome), sports_fan_funnel (conversion rates), state_sport_popularity (state shares), venue_capacity_utilisation (venue efficiency). Final submission use verified sources with proper citations.

Data sources including:

1. Australian Sports Commission — AusPlay Survey 2023-24
<https://www.clearinghouseforsport.gov.au/research/ausplay/results>
2. Australian Sports Commission — Annual Reports (High Performance Investment)
https://www.sportaus.gov.au/about/annual_reports
3. Australian Bureau of Statistics — Participation in Sport and Physical Recreation (cat. 4177.0)
<https://www.abs.gov.au/statistics/people/people-and-communities/participation-sport-and-physical-recreation>
4. Australian Olympic Committee — Medal Tallies (Tokyo 2020, Paris 2024)
<https://www.olympics.com.au/>
5. AFL Annual Report 2019-2024 — Total attendance figures
6. Cricket Australia Annual Report 2019-2024 — BBL and International attendance
7. NRL Annual Report 2019-2024 — Rugby League attendance
8. Sport Australia — Fan Engagement Survey 2024

HOW - VISUALISATION IDIOMS:

Eleven Vega-Lite visualisations with diverse idioms: (1) Choropleth map (Australian Albers projection) showing dominant sport by state with hover tooltips. (2-3) Heatmaps for male/female participation (rect marks, sequential colour scales). (4) Grouped bar chart comparing AFL participation by gender. (5) Stacked bar showing state sport shares (normalised). (6) Streamgraph for attendance trends (stacked area, centre-aligned). (7) Lollipop chart for venue utilisation (point + rule marks). (8) Slope chart comparing funding vs medals (faceted line chart). (9) Bubble chart for 3D analysis (circle size = funding). (10) Funnel chart showing fan conversion (bars with decreasing width simulation). (11) Alluvial-style bar matrix showing participation pathways. Interactivity: tooltips on all charts, responsive layout with Pure.css, typography using Playfair Display + Inter fonts, consistent colour schemes (AFL=red, Rugby=green, etc.), story structured in 5 acts with insight boxes. Custom-built elements include the funnel simulation and colour-coded geographic map. Total data <5MB for fast loading.

SPECIAL FEATURES:

- Single-scroll webpage, no horizontal scrolling on laptop
- Mobile-responsive using Pure.css grids
- AI acknowledgment: GitHub Copilot (code scaffolding), ChatGPT (text refinement)
- Metadata complete: data sources, author, date, AI statement in footer
- Sketch available in GitHub repository as PDF